master, but they do require some patience and a fair amount of index wrangling.

7.1 MATHEMATICAL INTERLUDE: TENSOR PRODUCTS IN COMPONENT FORM

In Lecture 6, we explained how to form the tensor product of two vector spaces using the abstract notation of bras, kets, and operator symbols like σ_z. How does that translate into columns, rows, and matrices?

Building tensor products from matrices and column vectors is not hard. The rules are straightforward, as we'll see below. The tricky part is understanding why these rules work—why they allow us to build matrices and column vectors *that have the properties we want*. We'll tackle the issue in two different ways. First, we'll build composite operators using the tried-and-true method we developed in Lecture 3. Then we'll show you how to build composite operators directly from their component operators.

7.1.1 Building Tensor Product Matrices from Basic Principles

Back in Lecture 3, we showed you how to write *any* observable **M** in matrix form, relative to a specific basis. Take a moment to review Eqs. 3.1 through 3.4. In that section, we calculated the numerical values m_{jk} of **M**'s matrix elements with the expression

$$m_{jk} = \langle j|\mathbf{M}|k\rangle, \tag{7.1}$$

where |*j*> and |*k*> represent the basis vectors. Each |*j*>, |*k*> combination generates a different matrix element.[1]

Our plan is to apply this formula to some tensor product operators and see what we get. Because of our double-indexing convention for tensor product basis vectors, the "sandwiches" in these equations will look a little different from the ones in Eq. 7.1. On each end of the sandwich, we will cycle through the basis vectors |*uu*>, |*ud*>, |*du*>, and |*dd*>.[2] To keep things simple, we'll use the operator $\sigma_z \otimes I$ as an example, where *I* is the identity operator.